

# NATHAN VARGHESE

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## EDUCATION

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### University of Colorado, Boulder

Expected Fall 2025

*Master of Science in Aerospace Engineering: Autonomous systems*

*Boulder, CO*

*Relevant coursework: Advanced robotics, Statistical estimation, Optimization, Big data*

### Maharashtra Institute of Technology (MIT ADT)

August 2018 - June 2022

*Bachelor of Technology in Aerospace Engineering, GPA 3.52*

*Pune, India*

## TECHNICAL SKILLS

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**Programming:** Python, ROS2, C++, MATLAB, JULIA, C#, Git, Linux

**Design and Simulation:** ROS-Unity, WeBots, AutoCAD, SolidWorks, ANSYS, Godot, Blender

## PROJECTS

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### Autonomous Race Car | *Python, ROS2*

December 2024

- Developed a fully autonomous racing car using a fine tuned Bug-Algorithm and PID for steering control for the AWS Deepracer that would go around a loop and our team won the first place in the class competition.
- Utilized GitHub for distributed workload and version control.

### Cooperative Air-Ground Robot Localization | *MATLAB, Python*

December 2023

- Validation of linearized and extended Kalman filters for accurate and efficient tracking for ASEN 5044 using Matlab.
- Derived the CT Jacobians and DT linearized model about the specified nominal point. Implemented the EKF and LKF to estimate state trajectories using observation data.[\(link\)](#)

### ROS-Unity Simulator for Open Motion Planning Library | *C++*

December 2023

- Setup of ROS-TCP connection over IP address for seamless connection with Unity physics simulator for ASEN 5254.
- Integrated the foundation of ROS communication utilizing python and C++ between Unity and planner using ROS topics and C# scripts and assigned to respective game objects.
- Used Sampling-based planners such as RRT\*, SST and PRM to find the optimal trajectory for problem space.[\(link\)](#)

## EXPERIENCE

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### University of Colorado Boulder

Boulder, CO

*Writing Consultant*

*August 2024 – Current*

- Collaborate with students across disciplines to enhance writing clarity, organization, and style through one-on-one consultations.
- Provide constructive feedback on academic essays, research papers, and creative projects, focusing on grammar, structure, and argument development.

*Graduate Teaching Assistant*

*August 2023 – Current*

- Assisted undergraduate students with various courses with office hours, grading and teaching.

### Department of Aerospace Engineering, MIT ADT

Pune, India

*Junior Research Fellow*

*November 2022 – April 2023*

- My team was awarded \$56,000 for the development of a rapid high speed apparatus for experimental material research from a renowned research establishment.
- Achieved a 12 fold increase in the frequency of testing at supersonic speeds ( $Mach > 1.2$ ) as compared to traditional methods of impact testing.
- Simulated the attachment for rotary components using finite element analysis (FEA) in ANSYS for Modal analysis, Factor of Safety, Stress analysis.

*Research Intern*

*May 2021 – August 2022*

- Supervised an 8-member fabrication team for over 400 hours. Managed \$50-60 weekly in vendor transactions.